Pratik Pranav

Computer Science & Engineering
Indian Institute of Technology, Delhi
pratkpranav

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portfolio
pratik-pranav

Academic Details

Year	Degree	Institute	CGPA/Percentage
2018-2022	B.Tech in Computer Science and Engineering	Indian Institute of Technology Delhi	8.226/10
2018	Class XII, CBSE	Keshav Saraswati Vidyamandir,Patna	89.6%
2016	Class X, CBSE	DAV Public School,Patna	10/10

Scholastic Achievements

- Received **Arjuna Award** for exceptional performance in 2022 and 2024 at ThirdAI.
- Secured **All India Rank 646** in Joint Entrance Exam Advanced - 2018 among 225,000 candidates
- Awarded **KVPY Fellowship** by Dept. of Science & Technology, Govt. of India
- Recipient of **National Talent Search Examination(NTSE)** Scholarship by NCERT, New Delhi
- Qualified **Regional Mathematics Olympiad(RMO)** twice in 2015-16 and 2016-17 organized by HBCSE

Publications

1. N. Meisburger, V. Lakshman, B. Geordie, J. Engels, D. Torres Ramos, **P. Pranav**, B. Coleman, B. Meisburger, S. Gupta, Y. Adunukota, S. Jain, T. Medini, A. Shrivastava, "BOLT: An Automated Deep Learning Framework for Training and Deploying Large-Scale Search and Recommendation Models on Commodity CPU Hardware," *Accepted at ACM International Conference on Information and Knowledge Management*, 2023. [Link]
2. A. Shrivastava, V. Lakshman, T. Medini, N. Meisburger, J. Engels, D. Torres Ramos, B. Geordie, **P. Pranav**, S. Gupta, Y. Adunukota, S. Jain, "From Research to Production: Towards Scalable and Sustainable Neural Recommendation Models on Commodity CPU Hardware," *Accepted at ACM Conference on Recommender Systems*, 2023. [Link]

Work Experience

Senior Software Engineer, ThirdAI — July 2024 - Present
AI Engineer, ThirdAI — May 2022 - June 2024

- **Enterprise Search Platform Development & Infrastructure:**
  - Architected and deployed a high-availability on-premises infrastructure (Nomad, Consul, Traefik) across 25-node clusters with automated health checks.
  - Developed an Ansible-driven, one-command installer for on-prem and multi-cloud (AWS, GCP, Azure) clusters.
  - Rolled out turnkey solutions to multiple Fortune 500 clients, ensuring seamless adoption and robust uptime.
  - Extended platform to cloud environments using Kubernetes and Terraform to enhance scalability, automation, and reliability.
  - Integrated Keycloak for secure SSO and fine-grained access control; maintained and optimized backend services for performance, fault tolerance, and seamless upgrades.
- **Led Development of ThirdAIs LLM Bolt2.5B:**
  - Pre-trained a 2.5 B-parameter generative LLM on 10 Intel Sapphire Rapids CPUs (112 cores), achieving 2 B tokens/day.
  - Employed dynamic sparsity algorithms and multiple architectural improvements to run efficiently under strict CPU/memory constraints while matching GPT-2-XL performance.
  - Published in Medium article, demos on GitHub.
- **Built Distributed Training Framework for ThirdAIs BoLT:**
  - Designed and led a Ray-based data-parallel training system for DLRM and other billion-parameter models on CPU clusters.
  - Integrated gradient compression to achieve 4.1x faster end-to-end convergence and 11x lower communication overhead versus full gradient training.
  - Extended framework usage across ThirdAI products (UDT, NeuralDB, Bolt, Generative Models).
  - Highlighted in AnyScale blog and reached top-5 on Hacker News (37325173).
- **Maintainer of ThirdAIs Internal ML Library:**
  - Contributed core features, performance optimizations, and rigorous test suites to ensure framework robustness.
  - Partnered with client teams to troubleshoot complex integration and deployment issues.
- **Mentoring & Onboarding of New Team Members:**
  - Mentored four new engineers through technical ramp-up, code reviews, and best practices.
  - Organized knowledge-sharing sessions and created onboarding documentation to accelerate productivity.

Internship

Summary Research Intern, University of Edinburgh April 2020 - July 2020
Under Prof. Antonio Barbalace, School of Informatics

- Implemented a messaging layer for multikernel communication using IVSHMEM and deployed it on QEMU.
- Utilized arm and x86 kernels and Popcorn Linux virtual machine images for deployment and testing.
- Studied interrupt transportation in a multi-kernel environment on QEMU to reduce latency.

Software Solution Intern, Samsung Electronics June 2021 - July 2021
Graphics Lab

- Designed an Augmented Reality-based PacMan game in Unreal Engine using ARCore.
- Developed an algorithm for calculating the working plane using ARCore Traced Planes, added movement for agents and ghosts in the game, and worked on world-building.

Course Projects

Portfolio Management System Prof. Abhilash Jindal, Jan 2022 - Mar 2022

- Made an web-application, which connects the popular crypto exchanges, stocks, download and store data for popular stocks
- Added support for visualizing historical data, their current portfolio, getting top quotes and perform analytical operations on the data
- Used data processing frameworks like Kafka, Spark, Redis and Cassandra and Node.js for backend

Simulating vehicles movements using Bayes and Kalman filter Prof. Rohan Paul, Feb 2021 - Mar 2021

- Implemented HMMs for filtering, smoothing and predicting the motion of an aerial vehicle in a simulated environment
- Implemented Kalman filter for estimating aerial motion using error-prone sensor data
- Used Hungarian Algorithm as data association strategy for filtering sensor data for multiple aerial vehicles observed simultaneously

Learning Optimal policy using MDP and Q-Learning Prof. Rohan Paul, Apr 2021 - May 2021

- Implemented Value Iteration and Q-Learning algorithm for a mobile robot movement to estimate optimal policy for its movement towards a goal state
- Observed dependence of discount factor, iterations on Value iteration and exploration parameter on Q-Learning algorithms

Metro Route Display Prof. Anshul Kumar, Sep 2019 - Oct 2019

- Displayed bus route between two user-specified locations on a CRT display with the help of a FPGA board
- Used Dijkstra algorithm to find the shortest path between two locations of the city

Technical Skills

Languages: Python, C++, C, Go, Java, OCaml, Prolog, Bash Scripting

Software: Docker, Kubernetes, Terraform, Nginx, FastAPI, Keycloak, PostgreSQL, SQLite, AWS, Azure, Google Cloud, Traefik, HashiCorp Nomad, HashiCorp Consul, HashiCorp Vault

Frameworks/Libraries: PyTorch, Ansible, Ray, GitHub, Hugging Face, OpenAI, Gloo

Relevant Courses

- **Computer Science:**
  Special Topics in AI: Planning and Estimation for Autonomous Systems, Introduction to AI, Machine Learning, Operating System, Cloud computing technology fundamentals, Parallel Programming and Distributed Systems, Introduction to Database Management Systems, Deep learning For mechanics, Analysis & Design of Algorithms, Computer Networks, Data Structures & Algorithms, Discrete Mathematics, Digital Logic & System Design, Programming Languages, Theory of Computation, Computer Architecture, Design Practices
- **Mathematics and Electrical Engineering:**
  Probability & Stochastic Processes, Graph Theory, Signals and Systems , Introduction to Elecrical Engineering, Linear Algebra, Calculus

Extra Curricular Activities

- Member of the **Algorithm and Coding Club**, IIT Delhi
- Taught children of orphanage under the program **Apna Parivar** of NSS IIT Delhi